

Comparison of a computed tomographic pulmonary trunk to aorta diameter ratio with echocardiographic indices of pulmonary hypertension in dogs

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Abstract

There are limited criteria for the detection of pulmonary hypertension in dogs undergoing computed tomography (CT) for pulmonary disease. This retrospective analytical exploratory study compared a CT pulmonary trunk to aorta ratio with echocardiographic estimates of pulmonary hypertension. Dogs having both a contrast thoracic CT and echocardiogram were selected and maximal pulmonary trunk and descending aorta diameters were measured by two observers on a single transverse CT image. Computed tomographic diameter ratios were compared with the echocardiographic parameters of tricuspid regurgitation gradient, right ventricular acceleration time-to-ejection time ratio, pulmonary insufficiency gradient, and pulmonary artery to aorta diameter. A total of 78 dogs were sampled, with 44 dogs having one or more finding suggestive of pulmonary hypertension. A moderate positive correlation was shown between tricuspid regurgitation gradient and CT pulmonary trunk to aorta ratio ($r = 0.61$, P -value < 0.0001). Mean CT pulmonary trunk to aorta ratio of dogs with moderate ($P = 0.0132$) and severe ($P < 0.0003$) pulmonary hypertension were significantly higher than normal dogs. There was no significant difference in mean CT pulmonary trunk to aorta ratio between normal and mild pulmonary hypertension dogs ($P = 0.4244$). The intraclass correlation coefficient (0.72) showed good reproducibility of the ratio. Findings indicated that CT pulmonary trunk to aorta ratio is a reproducible and potentially useful method to predict moderate and severe pulmonary hypertension in dogs, but not mild pulmonary hypertension. In dogs undergoing thoracic CT for pulmonary disease, an increased ratio should prompt follow up echocardiography.

KEYWORDS

canine, cardiovascular, CT, echocardiography

1 | INTRODUCTION

In veterinary medicine, thoracic helical computed tomography (CT) is an increasingly available and utilized modality for the evaluation of respiratory disease in dogs.^{1,2} Severe pulmonary parenchymal disease is a recognized cause of pulmonary hypertension and the ability to identify features of pulmonary hypertension using CT has the potential to modify patient management and improve clinical outcomes. The CT features of pulmonary hypertension have been characterized in people and quantitative measures are routinely used to aid in the diagnosis.³ These quantitative measures include an absolute measure of the pulmonary artery (equivalent to the pulmonary trunk in dogs) size and a ratio of the main pulmonary artery to aorta diameters (PA:Ao).^{4–6} In

some circumstances CT measurements have been reported to offer an improved diagnostic yield when compared with echocardiography in people.⁷ A CT ratio of this type has been evaluated in a small number of normal dogs, but has yet to be investigated in a larger population of dogs both with and without pulmonary hypertension.⁸

Pulmonary hypertension is a complex and serious clinical syndrome that results from primary, or more often, secondary dysfunction in the pulmonary vasculature. Pulmonary hypertension in dogs is defined as an elevated systolic pulmonary arterial pressure greater than approximately 30 mm Hg and has been documented in association with canine heartworm disease, left-sided cardiac disease, pulmonary parenchymal disease such as pulmonary fibrosis, pulmonary thromboembolic disease, or congenital cardiac conditions such as patent ductus

arteriosus or ventricular septal defect.^{9–14} Ideally, the diagnostic evaluation of pulmonary hypertension evaluates the pulmonary artery pressure and determines an underlying etiology. Right heart catheterization is the definitive technique to measure pulmonary artery pressure. However, cardiac catheterization is an invasive procedure that requires general anesthesia and is not routinely performed in dogs in a clinical setting.^{15–18} Two-dimensional and Doppler echocardiography currently provides the most accurate, noninvasive, available, and cost-effective estimation of pulmonary artery pressure in dogs. When tricuspid regurgitation is present an indirect estimation of pulmonary systolic arterial pressure can be obtained using the modified Bernoulli equation.^{15,18} Other parameters used to estimate the presence of pulmonary hypertension include evidence of right ventricular dilation and/or concentric hypertrophy, interventricular septal flattening, right atrial enlargement, the ratio of main pulmonary artery diameter to aortic root diameter, pulmonic valve diastolic pressure gradient (when insufficiency is present), pulmonary artery time to peak velocity or acceleration time, acceleration time to right ventricular ejection time ratio (AT:ET), right ventricular tissue Doppler imaging, tricuspid annular plane systolic excursion, and right pulmonary artery distensibility.^{12,15,17,19–24} Thoracic radiology is not considered a reliable test for pulmonary hypertension.⁹ However, radiographic abnormalities associated with pulmonary hypertension can include right-sided cardiac enlargement; generalized cardiomegaly; lobar pulmonary artery enlargement; abnormalities in pulmonary parenchyma; enlarged caudal vena cava; and hepatomegaly and ascites.^{9,25}

The aims of this study were to compare a CT pulmonary trunk to aorta ratio with commonly used echocardiographic estimations of pulmonary hypertension in dogs, including tricuspid regurgitation pressure gradient, right ventricular acceleration time-to-ejection time ratio, pulmonary insufficiency pressure gradient, and the ratio of main pulmonary artery diameter to aortic root diameter; and to evaluate the inter- and intraobserver variability of this CT ratio. Our hypotheses were that when compared with the CT pulmonary trunk to aorta ratio there would be a moderate to strong positive and negative correlation between the echocardiographic tricuspid regurgitation gradient and right ventricular acceleration time-to-ejection time ratio, respectively, and that the measurements of the CT pulmonary trunk to aorta ratio would have both low intraobserver and interobserver variability.

2 | MATERIALS AND METHODS

2.1 | Patient selection

For this retrospective analytical exploratory study, the electronic medical record and the radiology picture archiving and communication system (PACS) were evaluated for all dogs that had both a thoracic CT with contrast and echocardiography between 2008 and 2015. The CT and echocardiography studies needed to be performed within 1 week of each other to minimize the likelihood of a significant change in disease severity between the tests. The machine type, scan parameters,

and contrast agent used for the CT acquisition were noted from the records. Study inclusion was independent of clinical disease state or the presence or absence of pulmonary hypertension on echocardiography. All decisions for subject inclusion or exclusion were made by the consensus of two observers (E.J.H. and J.S.S.).

2.2 | Echocardiography

Echocardiographic still images and cine-loops, including standard M-mode, two-dimensional, color-flow, and spectral Doppler were retrospectively reviewed on a cardiology workstation (GE EchoPAC, GE Healthcare, Waukesha, WI) by a board-certified cardiologist (S.M.C.). The cardiologist was not aware of CT or clinical findings at the time of image review. Ultrasound machine specifications and scan protocols were noted from the image data. The end-diastolic diameter of the pulmonary trunk, referred to more commonly as the pulmonary artery on echocardiographic studies, just beneath the closed pulmonic valve leaflets, and the aortic root dimension measurements were reviewed using the right parasternal short-axis view, allowing the calculation of the main pulmonary artery diameter to aortic root diameter ratio, as previously described.¹² Pulmonic stenosis was excluded on the basis of a normal pulmonic valve ejection velocity; absence of abnormal doming and thickening of the pulmonic valve leaflets; and lack of poststenotic dilation of the pulmonary artery.¹⁵

Color-flow Doppler images and cine-loops of the tricuspid valve obtained from the right parasternal long-axis and left apical four-chamber views were reviewed. In the presence of tricuspid regurgitation, the peak instantaneous velocity was measured utilizing continuous wave Doppler through the regurgitant jet. The modified Bernoulli equation ($\Delta P = 4 \times \text{velocity}^2$) was utilized automatically by the ultrasound machine software to estimate the maximal systolic pressure gradient between the right ventricle and the right atrium.

Pulmonary artery flow profiles were evaluated from the right parasternal short-axis view at the level of the heart base. The pulsed wave sample volume was placed at the level of the pulmonic valve leaflets, in the center of the pulmonary artery. The peak velocity was recorded and the presence or absence of mid-systolic notching of the pulmonary artery flow profile was noted. The right ventricular acceleration time was measured as the time from onset of pulmonary flow to the peak flow velocity; the ejection time was calculated as the time from beginning to end of pulmonic flow; and the right ventricular acceleration time-to-ejection time ratio was calculated, as previously described.¹⁹ The velocity of pulmonic insufficiency, when present, was measured in the right parasternal short-axis view at the level of the heart base via continuous wave Doppler, and the ultrasound software automatically calculated the maximal diastolic pressure gradient between the right ventricle and the pulmonary artery using a modified Bernoulli equation.

A diagnosis of pulmonary hypertension was made based on a peak tricuspid regurgitation pressure gradient of greater than 30 mm Hg; a peak pulmonary insufficiency pressure of greater than 16 mm Hg;¹² a ratio of main pulmonary artery diameter to aortic root diameter ratio >0.98 ;¹⁷ and/or an right ventricular acceleration time-to-ejection time ratio of less than 0.31.¹⁹ Four categories of peak

tricuspid regurgitation pressure gradient were utilized based on previously described thresholds of severity: no pulmonary hypertension, tricuspid regurgitation ≤ 30 mm Hg; mild pulmonary hypertension, tricuspid regurgitation $31\text{--}\leq 50$ mm Hg; moderate pulmonary hypertension, tricuspid regurgitation $51\text{--}\leq 75$ mm Hg; and severe pulmonary hypertension, tricuspid regurgitation > 75 mm Hg, respectively.¹²

2.3 | Computed tomography

The postcontrast transverse CT images were independently reviewed by two board-certified radiologists (A.S., J.S.S.) on dedicated PACS Workstations (Carestream Vue PACS, Carestream, Rochester, NY). The radiologists were not aware of the dog's echocardiographic or clinical diagnosis. When there was a choice, the soft tissue algorithm images were utilized and all images were viewed in a soft tissue (mediastinal) window (level 60, window width 360). Arterial phase angiogram images were used for the measurements in preference to the delayed contrast images, when available. The transverse image was selected where the pulmonary trunk diameter was widest. The electronic calipers were placed on either side of the contrast-filled lumen at the widest diameter perpendicular to the longitudinal axis of the vessel, just ventral to where it flared and branched into the left and right pulmonary arteries. On the same image, the short-axis dimension of the descending aorta was measured. The angle of this short-axis measurement was guided by evaluating the curvature of the aorta on the images both cranial and caudal to the measured images and by evaluating the shortest dimension, when an ovoid image of the aorta was present. The descending part of the aortic arch was chosen over the ascending part, as in our population, the ascending aortic margins were often poorly defined. The same image was used to measure both the aorta and pulmonary trunk ensures that both vessels were in the same phase of the cardiac cycle (Fig. 1). As they were not uniformly available for all cases, no multiplanar reconstructions were used to obtain the vessel measurements. Both observers obtained the measurements on two different occasions. A separation of at least 1 week was required before each observer repeated a measurement on the same study.

2.4 | Statistical analysis

The echocardiographic and CT measurements were tabulated on an excel worksheet and transferred to statistical software (Statistical Analysis Software, SAS Institute, Cary, NC. Version 9.2) for analysis by a biostatistician (B.A.B.). Echocardiographic variables evaluated included peak tricuspid regurgitation pressure gradient, the right ventricular acceleration time-to-ejection time ratio, peak pulmonary insufficiency pressure gradient, and the two-dimensional ratio of the main pulmonary artery to aorta. Computed tomography variables included the two measurements of the pulmonary trunk to aorta ratios at different timepoints from each of the two observers and the mean of these four pulmonary trunk to aorta ratio measurements. A *P*-value of <0.05 was utilized as the threshold of statistical significance.

The correlation between the mean CT pulmonary trunk to aorta ratio and the four echocardiographic parameters as follows: tricuspid

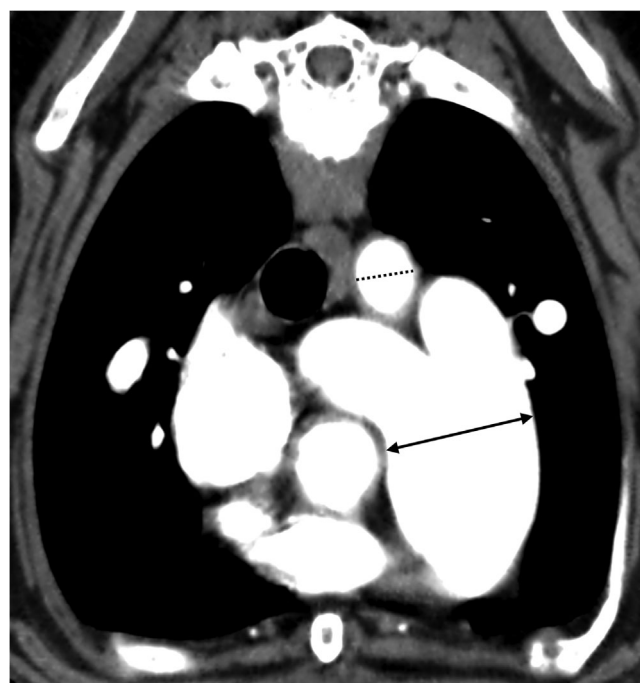


FIGURE 1 Arterial-phase postcontrast transverse thoracic angiogram computed tomographic (CT) image at the level of the pulmonary trunk in a 23 kg Siberian Husky dog with an estimated tricuspid regurgitation pressure gradient of 83 mm Hg on an echocardiogram. The CT pulmonary trunk to aorta ratio measurement technique consists of the maximal pulmonary trunk diameter measured immediately ventral to the bifurcation into the left and right pulmonary arteries (solid line with arrowheads; mean 29.6 mm) and the short axis of the descending part of the aorta diameter (dashed line, mean 11.6 mm). The mean measured CT pulmonary trunk to aorta ratio in this dog was 2.55

regurgitation pressure gradient, right ventricular acceleration time-to-ejection time ratio, pulmonary insufficiency gradient, and the ratio of main pulmonary artery diameter to aortic root diameter were evaluated using the Pearson's correlation coefficient. The normal distribution of the data was evaluated graphically. The strength of the correlations was reported both numerically and qualitatively using previously reported guidelines.²⁶

The means, standard deviations, minima, and maxima of the CT pulmonary trunk to aorta ratio for the four categories of pulmonary hypertension severity as determined by tricuspid regurgitation gradient were calculated. An analysis of variance (ANOVA) model was used to perform an F-test that evaluated for a difference in group means, which was followed Dunnett's procedure that evaluated for significant mean differences between each disease severity level and the control group.

The intraobserver and interobserver variability of the CT pulmonary trunk to aorta ratio between the two radiologists and the two timepoints was assessed using both the coefficient of variation (CV) and the intraclass correlation coefficient (ICC). For the intraclass correlation coefficient, a mixed effect model of the variable CT pulmonary trunk to aorta was utilized and a previously reported scale of an ICC < 0.4 indicating poor reproducibility, $0.4 < \text{ICC} < 0.75$ indicates

fair to good reproducibility, and ICC ≥ 0.75 indicating excellent reproducibility was utilized.²⁷

3 | RESULTS

There were 78 dogs that met the inclusion criteria and no cases were excluded due to an absence of CT or echocardiographic images. The median time between echocardiography and CT was 1 day (range: 0–7 days). Thirteen CT studies were obtained using a fourth-generation single detector row helical scanner (PQ-5000, Picker Corporation, Cleveland, OH) and 65 were obtained using a third-generation 16-detector row helical scanner (Aquilion-16, Toshiba Medical Systems, Tochigi, Japan). Scan parameters were customized for the individual patients, although generally they were divided into two major protocols as follows: studies performed with the single detector row scanner were helical volumetric acquisitions reconstructed into contiguous transverse 5-mm thick images using a lung algorithm. The scans were performed both before and after intravenous injection of either iohexol (Omnipaque 300, GE Healthcare, Princeton, NJ) at a dose of 600 mg/kg or diatrizoate meglumine (Hypaque, GE Healthcare, Princeton, NJ) at a dose of 564 mg/kg. The standard protocol for dogs undergoing pulmonary artery angiography was to use a test bolus using a one-fourth dose of contrast to establish the image acquisition timing. The multidetector row studies were helical volumes reconstructed into transverse, sagittal, and dorsal plane images using algorithms for both lung and soft tissue. Contiguous 3- or 5-mm thick images were reconstructed with the later typically being used with patients weighing greater than 25 kg. Bolus tracking software was used for the multidetector row angiography studies. A region of interest placed in the pulmonary trunk and the scan acquisition was triggered by a Hounsfield unit value of >150 . For studies obtained with both scanners, delayed postcontrast images were typically acquired 1–2 min following contrast administration. Arterial phase postcontrast images were available in 42 dogs and only delayed postcontrast images were available in 36 dogs. Hyperventilation induced apnea and positive pressure ventilation at 10 mm Hg were a standard part of the thoracic CT protocol during the study period. No electrocardiogram gating was utilized.

The echocardiograms were either performed by or supervised by a board-certified cardiologist. The ultrasound machine (GE Vivid 7™, GE Healthcare, Waukesha, WI) was equipped with 3, 7.5 and 10 MHz variable frequency phased array transducers. Standard 2D guided M-mode measurements were available from the right parasternal short-axis view, obtained according to the recommendations of the American Society of Echocardiography.²⁸ A total of 32 dogs had one or more of the following indicators of pulmonary hypertension: abnormal tricuspid regurgitation gradient, right ventricular acceleration time-to-ejection time ratio, or pulmonary insufficiency gradient suggestive of pulmonary hypertension. Seven dogs had both an abnormal tricuspid regurgitation gradient and a decreased right ventricular acceleration time-to-ejection time ratio. An additional 12 dogs had an echocardiographic main pulmonary artery diameter to aortic root diameter ratio that exceeded 0.98 as their only echocardiographic

abnormality. This total population included 19 neutered males, 20 neutered females, 2 intact males, and 3 intact females. The median age at the time of echocardiography was 11.2 years (range: 3.6–15.0 years). The median weight of the dogs was 10.6 kg (range: 2.8–40.0 kg). The included breeds were as follows: Chihuahua (4/44), Golden Retriever (3/44), Mixed breed (3/44), Bichon Frise (2/44), Cocker Spaniel (2/44), Miniature Pinscher (2/44), Pug (2/44), Shetland Sheepdog (2/44), and one each of American Pit Bull Terrier, Beagle, Bernese Mountain Dog, Boston Terrier, Brittany Spaniel, Cairn Terrier, Cavalier King Charles Spaniel, Cocker Spaniel, Coonhound, German Shepherd, Italian Greyhound, Jack Russell Terrier, Jindo, Maltese, Miniature Poodle, Miniature Schnauzer, Pomeranian, Portuguese Water Dog, Samoyed, Sealyham Terrier, Siberian Husky, Soft Coated Wheaten Terrier, West Highland White Terrier, and Yorkshire Terrier.

The population of 34 dogs with no echocardiographic indicators of pulmonary hypertension included 18 neutered males, 10 neutered females, 4 intact males, and 2 intact females. The median age at the time of echocardiography was 6.8 years (range: 1.6–13.1 years). The median weight of the dogs was 15.25 kg (range: 6.4–77.2 kg). The included breeds were as follows: Beagle (6/34), Mixed breed (5/34), Golden Retriever (4/34), Labrador Retriever (3/78), Shih Tzu (2/34), Basset Hound, Boxer, Cocker Spaniel, Collie, Dachshund, English Bulldog, French Bulldog, Italian Greyhound, Lhasa Apso, Mastiff, Miniature Schnauzer, Newfoundland, Shiba Inu, and Weimaraner.

Peak tricuspid regurgitation gradients (mm Hg) could be measured in 39/49 dogs with tricuspid regurgitation, with a median value of 36 mm Hg (range: 21–91 mm Hg). Thirteen dogs had normal tricuspid regurgitation gradient (pressure ≤ 30 mm Hg) while 26/39 dogs had a tricuspid regurgitation gradient >30 mm Hg, suggestive of systolic pulmonary hypertension. Further analysis of the tricuspid regurgitation gradients demonstrated that 12/39 dogs had mild pulmonary hypertension (31– ≤ 50 mm Hg), 8/39 had moderate pulmonary hypertension (51– ≤ 75 mm Hg), and 6/39 had severe pulmonary hypertension (>75 mm Hg). The ratio of right ventricular acceleration time-to-ejection time was measured in 53 dogs, there were 40 dogs that had a normal ratio ≥ 0.31 (median: 0.4 range: 0.31–0.57) and 13/53 dogs had a ratio <0.31 suggestive of pulmonary hypertension. Six dogs had a right ventricular acceleration time-to-ejection time ratio <0.31 that did not have a measurable tricuspid regurgitation gradient. Five of the dogs with a right ventricular acceleration time-to-ejection time ratio <0.31 had “notching” of the pulmonic velocity profile, suggestive of severe elevations in pulmonary systolic pressure. The main pulmonary artery diameter to aortic root diameter ratio was measured on the echocardiograms of 75/78 dogs with 38/75 dogs having a main pulmonary artery diameter to aortic root diameter ratio >0.98 . Pulmonic insufficiency was measured in 57 dogs with a median pressure gradient of 3.5 mm Hg (range: 1.4–63 mm Hg). An elevated pulmonary insufficiency pressure gradient (>16 mm Hg) was identified in two dogs, which had gradients of 50 and 63 mm Hg, respectively.

Pearson's correlation coefficients (Fig. 2) demonstrated statistically significant moderate linear correlation between the tricuspid

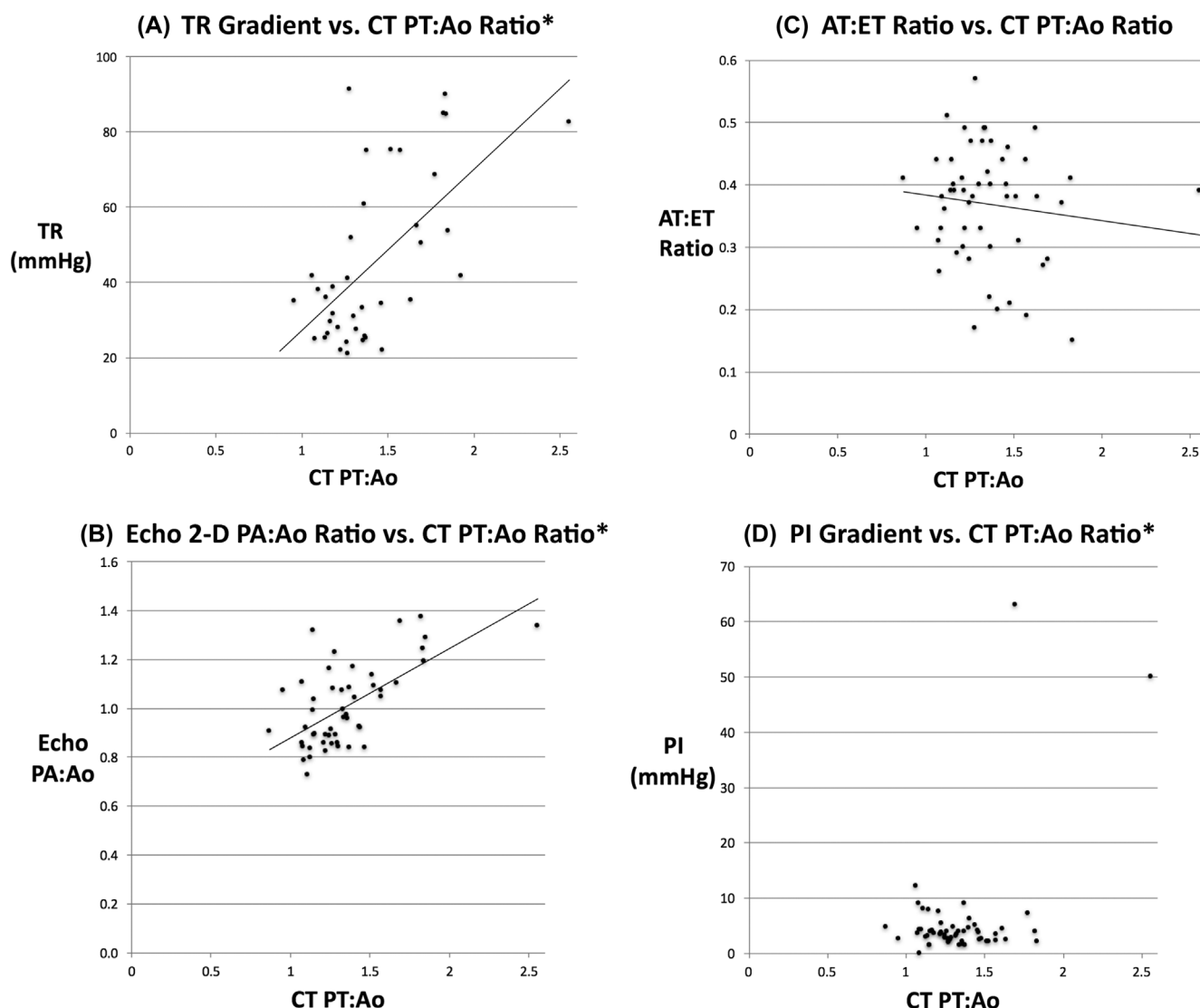


FIGURE 2 Graphical scatterplots displaying the four echocardiographic parameters that were correlated with the computed tomography pulmonary trunk to aorta diameter ratio (CT PT:Ao). A, The tricuspid regurgitation (TR) gradient demonstrated a statistically significant moderate correlation with the CT pulmonary trunk to aorta ratio ($r = 0.61$, P -value < 0.0001). B, The echocardiographic measurement of two-dimensional pulmonary artery to aortic diameter ratio (2-D PA:Ao) showed a weak, almost moderate positive correlation with the CT pulmonary trunk to aorta ratio ($r = 0.49$, P -value < 0.0001). C, The right ventricular acceleration time-to-ejection time ratio (AT:ET) was negatively associated with the CT pulmonary trunk to aorta ratio, however this relationship exhibited a very low negative correlation coefficient ($r = -0.12$) that failed to reach statistical significance (P -value = 0.40). D, The pulmonic insufficiency (PI) pressure gradient showed a weak, almost moderate positive correlation with the CT pulmonary trunk to aorta ratio ($r = 0.49$, P -value < 0.0001). It should be noted that the two high-leverage data points on scatter plot D represent the only two dogs with an abnormal measured pulmonary insufficiency gradient of 50 and 63 mm Hg, respectively. The * symbols denote a statistically significant correlation

regurgitation gradients and mean CT pulmonary trunk to aorta ratio ($n = 39$, $r = 0.61$, P -value < 0.0001). As anticipated, the right ventricular acceleration time-to-ejection time ratio was negatively associated with the pulmonary trunk to aorta ratio, however this relationship exhibited a very low negative correlation coefficient ($n = 53$, $r = -0.12$) that failed to reach statistical significance (P -value: 0.40). Given this was an unexpected result, we evaluated the correlation of the right ventricular acceleration time-to-ejection time ratio to the tricuspid regurgitation gradient, which showed a moderate and significant negative correlation ($n = 30$, $r = -0.57$, P -value: 0.0009). The correlation of the echocardiographic measurement of main pulmonary

artery diameter to aortic root diameter ratio showed a weak, almost moderate positive correlation with the CT pulmonary trunk to aorta ratio ($n = 75$, $r = 0.49$, P -value < 0.0001). The pulmonary insufficiency gradient showed a weak, almost moderate positive correlation with the CT pulmonary trunk to aorta ratio ($n = 59$, $r = 0.49$, P -value < 0.0001).

When evaluated across the dataset, the results of the F-test from the analysis of variance model indicated a significant difference in the mean CT pulmonary trunk to aorta ratios ($p < 0.0012$) between the four severity categories of pulmonary hypertension derived from the tricuspid regurgitation gradients. When evaluating each

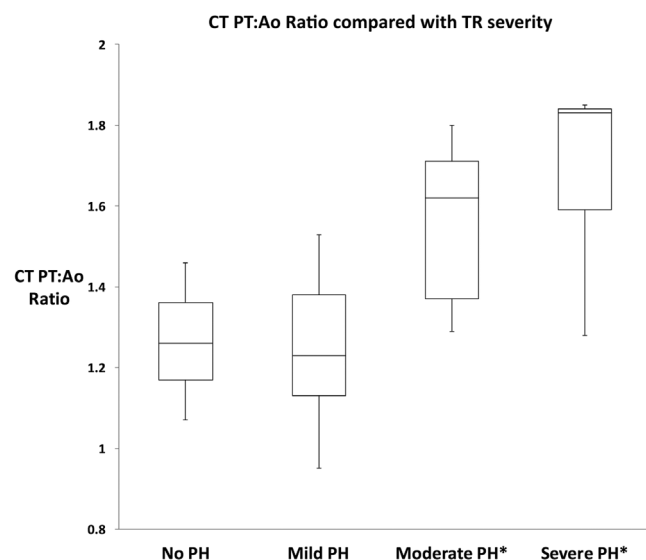


FIGURE 3 Box and Whisker Plot of the distribution of mean measured computed tomographic (CT) pulmonary trunk to aorta ratio values (y-axis) grouped by the four categories of pulmonary hypertension severity derived from the following tricuspid regurgitation pressure gradients: no pulmonary hypertension ($n = 13$), tricuspid regurgitation ≤ 30 mm Hg; mild pulmonary hypertension ($n = 12$), tricuspid regurgitation $31 - \leq 50$ mm Hg; moderate pulmonary hypertension ($n = 8$), tricuspid regurgitation $51 - \leq 75$ mm Hg; and severe pulmonary hypertension ($n = 6$), tricuspid regurgitation > 75 mm Hg (x-axis). A statistical difference (*) was noted between the dogs without pulmonary hypertension and those with both moderate and severe pulmonary hypertension

category independently the CT pulmonary trunk to aorta ratio of dogs with moderate (estimated difference = 0.16, $P = 0.0132$) and severe (estimated difference = 0.28, $P = 0.0003$) pulmonary hypertension were significantly higher than those with normal tricuspid regurgitation gradients (Fig. 3 and Table 1). There was no statistically significant difference in pulmonary trunk to aorta between the dogs with mild pulmonary hypertension and no pulmonary hypertension ($P = 0.4244$).

The intraclass correlation coefficient for the CT pulmonary trunk to aorta ratio was 0.72, which is at the high end of the fair to good range, suggestive of good interobserver reproducibility. The coefficients of variation (Table 2) also showed acceptably low intraobserver variability.

TABLE 1 Differences in the mean pulmonary trunk to aorta ratio grouped by the different severities of pulmonary hypertension based on echocardiography tricuspid regurgitation gradients

	<i>n</i>	Mean CT PT:Ao (Std dev; min-max)	ANOVA ratio difference from no PH	
			Estimated difference	<i>P</i> -value
No PH ≤ 30 mm Hg	13	1.26 (0.11; 1.07-1.47)	N/A	
Mild PH $31 - \leq 50$ mm Hg	12	1.30 (0.27; 0.95-1.92)	0.05	0.4244
Moderate PH $51 - \leq 75$ mm Hg	8	1.57 (0.20; 1.29-1.85)	0.16	0.0132*
Severe PH > 75 mm Hg	6	1.81 (0.43; 1.28-2.55)	0.28	0.0003*

Notes. The estimated difference is an output from an analysis of variance Dunnett's t-test that evaluated the difference in the groups means when compared with the control group (dogs with no pulmonary hypertension). ANOVA, analysis of variance; CT PT:Ao, computed tomography pulmonary trunk to aorta diameter ratio; min-max, minimum-maximum; PH, pulmonary hypertension; Std dev, standard deviation.

*Statistically significance difference compared to no PH.

TABLE 2 Coefficient of variation of the computed tomography pulmonary trunk to aorta ratio as a percentage between two radiologists and between the two different observation time periods of each radiologist

Radiologist	Interpretation	Mean CT PT:Ao ratio	SD CT PT:Ao ratio	Coefficient of variation
1	A	1.36	0.27	20.12
	B	1.41	0.32	22.43
	Mean	1.39	0.30	21.40
2	A	1.31	0.27	20.48
	B	1.33	0.31	23.17
	Mean	1.32	0.29	21.83

CT PT:Ao, computed tomography pulmonary trunk to aorta diameter ratio; SD, standard deviation; Interpretation A, first measurement of each dog by each radiologist; Interpretation B, second measurement of each dog by each radiologist.

4 | DISCUSSION

When compared with common echocardiographic indices, the CT pulmonary trunk to aorta ratio measured on a single transverse image may be a useful additional technique to aid in the detection of moderate and severe pulmonary hypertension. This ratio may be most clinically applicable in dogs that undergo thoracic CT for pulmonary disease. If an elevated ratio is found, these dogs may benefit from further evaluation with echocardiography. The methodology is straightforward and reproducible, with two easily defined anatomic landmarks that can be measured on routine transverse CT images.

Our investigation demonstrated a moderate correlation between tricuspid regurgitation gradient and the CT pulmonary trunk to aorta ratio measured on postcontrast CT images. Dogs with moderate and severe pulmonary hypertension had CT pulmonary trunk to aorta ratios significantly higher than both normal dogs and dogs with mild pulmonary hypertension. There was a high degree of overlap between the CT pulmonary trunk to aorta ratios of the no pulmonary hypertension and mild pulmonary hypertension groups with no statistically significant difference identified. While this result was unexpected, it has been suggested from other studies that some structural changes associated with pulmonary hypertension, such as right ventricular enlargement are not always present and usually occur in moderate to severe cases, a phenomenon that may extend to the dilation pattern

of the main pulmonary.^{12,15} This idea may be supported by a study of echocardiographic main pulmonary artery diameter to aortic root diameter ratio and tricuspid regurgitation gradient where there was no statistical difference between the control and mildly affected groups.²²

As shown in results (Table 1) the mean CT pulmonary trunk to aorta ratio ($\pm$ standard deviation) of the dogs with normal tricuspid regurgitation gradients was 1.26 ($\pm$ 0.11) and is higher than previously reported in 10 normal middle-sized to large breed dogs 1.11 ($\pm$ 0.15).⁸ These differences may be explained by a different aortic measurement technique with our study using the descending aortic arch rather than the ascending, or an inherent variability of the two relatively small canine populations. The descending aorta measurement was chosen for our study due to concerns that accurate measurement of the ascending part of the aorta may not be possible in all dogs as the result of motion artifact and poor demarcation of the vessel margins. In comparison, we anticipated that descending aorta would have consistently well-defined margins. A study comparing the two measurement techniques will be necessary to determine the best approach to evaluate the presence of pulmonary hypertension in dogs. An optimal CT pulmonary trunk to aorta ratio threshold for the diagnosis of pulmonary hypertension was not determined from this study and would be best evaluated by a follow-up prospective study. When evaluating the distribution of our data (Fig. 3) a CT pulmonary trunk to aorta ratio threshold of approximately 1.4 may be useful to identify cases more likely to have moderate or severe pulmonary hypertension. The established mean ratio thresholds used in adult people (>1.1)⁶ and ($>0.9:1$)⁴ and in human pediatrics (>1.09)²⁹ are lower than the normal CT pulmonary trunk to aorta ratio values reported in dogs, which is not unexpected given the differences in the cross-sectional anatomy between dogs and people, and particularly in our study where the smaller descending part of the aorta was measured on the transverse image.

There was good intraobserver and interobserver agreement between the two radiologists for the CT pulmonary trunk to aorta ratio measurements. The intraclass correlation coefficient and coefficient of variation both demonstrated good agreement both between the observers and between the two observation time periods, suggesting that the technique should be able to be used by other clinicians.

In the absence of discernable or accurate tricuspid regurgitation gradient, the right ventricular acceleration time-to-ejection time ratio has been proposed as a means to estimate systolic pulmonary artery pressures in both dogs^{19,22} and people³⁰. In our study, the acceleration time-to-ejection time ratio was measured in 53 dogs and 13 dogs had a ratio <0.31 , compatible with pulmonary hypertension. However, the right ventricular acceleration time-to-ejection time ratio was poorly correlated to the CT pulmonary trunk to aorta ratio with no statistical significance. The cause for this finding was uncertain given the moderate correlation, we observed between tricuspid regurgitation and the right ventricular acceleration time-to-ejection time ratio (-0.57 , $P = 0.0009$) that was similar to results published in other echocardiographic studies.^{19,22}

Additional echocardiographic techniques not utilized in this study have been evaluated and show promise for the noninvasive diagnosis

of pulmonary hypertension in dogs. These techniques include right ventricular tissue Doppler imaging,¹⁷ and measurement of tricuspid annular plane systolic excursion (TAPSE)²³ and right pulmonary artery distensibility.²² These additional noninvasive techniques may be useful to further validate the CT pulmonary trunk to aorta ratio prospectively against a wider range of echocardiographic parameters.

Some limitations in our study included the retrospective use of a largely client-owned population of dogs where we relied on non-invasive echocardiographic estimations of pulmonary hypertension. In human medicine, pulmonary pressures are routinely measured via direct right heart catheterization, which has superior sensitivity and accuracy and its use would likely have aided the evaluation of the CT pulmonary trunk to aorta ratio in dogs. However, in our clinical veterinary patients this catheterization technique requires general anesthesia, is invasive and expensive and therefore is rarely considered practical. In an experimental canine model of acute pulmonary hypertension, noninvasive echocardiographic measures of pulmonary arterial pressure have been shown to be imperfect, but moderately correlated with catheter pulmonary artery pressure measurements.³¹ The possibility of measurement errors arising from physiologic differences between conscious echocardiographic studies and CT performed under general anesthesia also need to be considered. Similarly, although there was median time separation of only 1 day between the echocardiogram and CT, and all dogs were required to have the studies completed within a week of each other, there is also the possibility that the pulmonary artery pressure varied from a variation in disease state during the interval between the two studies. The timing and delivery rate of the contrast relative to the scan acquisition was not standardized in this retrospective population. The effects of different contrast concentrations within the vessels at different phases of an angiogram may have affected their measured dimension. A standardized contrast administration protocol will aid prospective evaluation of this technique. Transverse images were utilized for all measurements as this was the standard archived image plane across the study population. While the measurement technique tried to minimize the risk of measuring geometric distortions of the vessels, the use of reformatted images oriented transverse to the pulmonary trunk and aortic arch may have yielded accurate measurement data and should be considered to validate the use of the transverse plane during further prospective studies. Finally, we were hampered by relatively small group sizes and may have lacked the statistical power to fully evaluate some of the relationships. Similarly, although the observer variability was low, it may have contributed to the inability to detect small CT pulmonary trunk to aorta ratio differences between groups.

In conclusion, the CT pulmonary trunk to aorta ratio may be useful for the detection of moderate and severe pulmonary hypertension in dogs. However, the CT pulmonary trunk to aorta ratio is not useful in the detection of mild pulmonary hypertension. We anticipate that the CT pulmonary trunk to aorta ratio, when used in dogs undergoing thoracic CT, has the potential to increase the use of follow-up echocardiography, and therefore expand the detection and treatment of pulmonary hypertension in clinical practice.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- (a) Conception and Design: Hankin E, Sutherland-Smith J, Sato A, Cunningham S, Barton B
- (b) Acquisition of Data: Hankin E, Sutherland-Smith J, Sato A, Cunningham S
- (c) Analysis and Interpretation of Data: Sutherland-Smith J, Hankin E, Sato A, Barton B

Category 2

- (a) Drafting the Article: Hankin E, Sutherland-Smith J, Cunningham S
- (b) Revising Article for Intellectual Content: Hankin E, Sutherland-Smith J, Sato A, Cunningham S, Barton B

Category 3

- (a) Final Approval of the Completed Article: Hankin E, Sutherland-Smith J, Sato A, Cunningham S, Barton B

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How to cite this article: Sutherland-Smith J, Hankin EJ, Cunningham SM, Sato AF, Barton BA. Comparison of a computed tomographic pulmonary trunk to aorta diameter ratio with echocardiographic indices of pulmonary hypertension in dogs. *Vet Radiol Ultrasound*. 2018;59:18–26. <https://doi.org/10.1111/vru.12540>